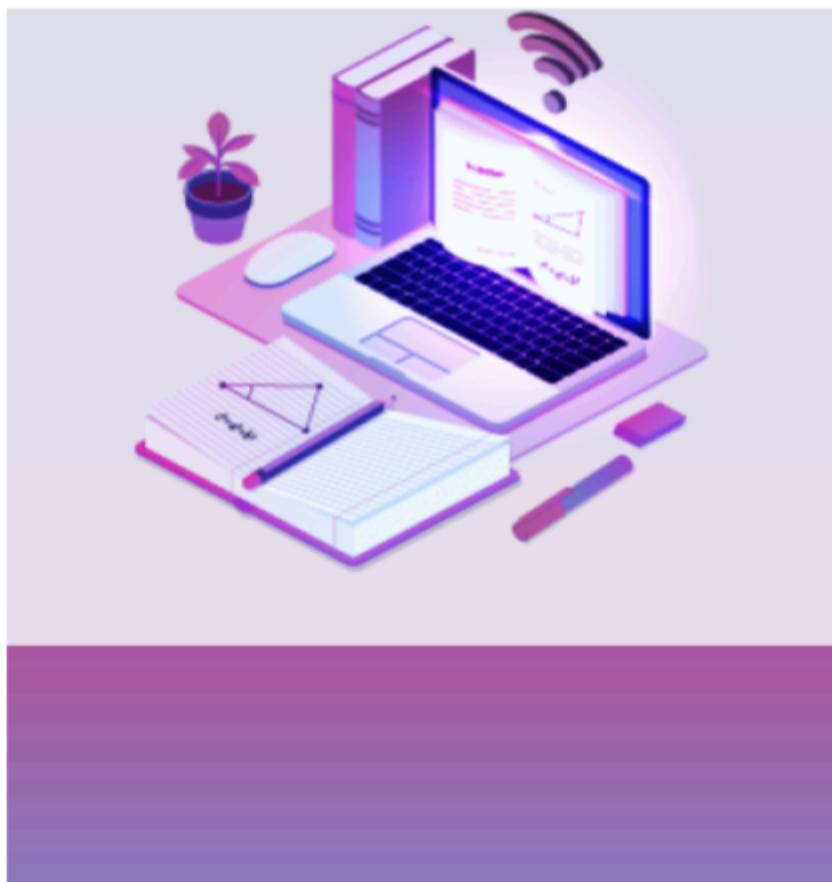


CAT Quadratic Equations Formulas PDF

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CAT Quadratic Equations

Formulas

- Quadratic Equations is also an important topic For CAT Exam.
- The theory involved in this topic is very simple and students should be comfortable with some basic formulas and concepts.
- The techniques like option elimination, value assumption can help to solve questions from this topic quickly.
- This pdf covers all the important formulas and concepts related to Quadratic Equations.
- General Quadratic equation will be in the form of $ax^2 + bx + c = 0$

- The values of 'x' satisfying the equation are called roots of the equation.
- The value of roots, p and q = $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- The above formula is known as the Shreedhara Acharya's Formula, after the ancient Indian Mathematician who derived it.
- Sum of the roots = $p+q = \frac{-b}{a}$
- Product of the roots = $p \times q = \frac{c}{a}$
- If 'c' and 'a' are equal then the roots are reciprocal to each other.
- If $b = 0$, then the roots are equal and are opposite in sign.



Let D denote the discriminant, $D = b^2 - 4ac$.

Depending on the sign and value of D , nature of the roots would be as follows:

- $D < 0$ and $|D|$ is not a perfect square:

Roots will be in the form of ' $p+iq$ ' and ' $p-iq$ ' where p and q are the real and imaginary parts of the complex roots. p is rational and q is irrational.

- $D < 0$ and $|D|$ is a perfect square:

Roots will be in the form of $p+iq$ and $p-iq$ where p and q are both rational.

- $D = 0 \Rightarrow$ Roots are real and equal $\left\{x = \frac{-b}{2a}\right\}$

- $D > 0$ and D is not a perfect square:

Roots are conjugate surds of the form $p + \sqrt{q}$ and $p - \sqrt{q}$

- $D > 0$ and D is a perfect square:

Roots are real, rational and unequal

→ **Signs of the roots:** Let P be product of roots and S be their sum

- $P > 0, S > 0$: Both roots are positive
- $P > 0, S < 0$: Both roots are negative
- $P < 0, S > 0$: Numerical smaller root is negative and the other root is positive
- $P < 0, S < 0$: Numerical larger root is negative and the other root is positive

Minimum and maximum values of $ax^2 + bx + c = 0$

- If $a > 0$: minimum value $= \frac{4ac - b^2}{4a}$ and occurs at $x = \frac{-b}{2a}$
- If $a < 0$: maximum value $= \frac{4ac - b^2}{4a}$ and occurs at $x = \frac{-b}{2a}$



If $A_n X^n + A_{n-1} X^{n-1} + \dots + A_1 X + A_0$, then

- Sum of the roots = $\frac{(-1)A_{n-1}}{A_n}$
- Sum of roots taken two at a time = $\frac{A_{n-2}}{A_n}$
- Sum of roots taken three at a time = $\frac{(-1)A_{n-3}}{A_n}$ and so

on Product of the roots = $\frac{[(-1)^n A_0]}{A_n}$

Finding a Quadratic Equation:

- If roots are given:

$$(x - a)(x - b) = 0 \Rightarrow x^2 - (a + b)x + ab = 0$$

- If sum s and product p of roots are given:

$$x^2 - sx + p = 0$$

- If roots are reciprocals of roots of equation

$$ax^2 + bx + c = 0, \text{ then equation is}$$

$$cx^2 + bx + a = 0$$

- If roots are k more than roots of $ax^2 + bx + c = 0$ then equation is $a(x - k)^2 + b(x - k) + c = 0$
 - If roots are k times roots of $ax^2 + bx + c = 0$ then equation is $a(x/k)^2 + b(x/k) + c = 0$
 - Descartes Rules: A polynomial equation with n sign changes can have a maximum of n positive roots. To find the maximum possible number of negative roots, find the number of positive roots of $f(-x)$.
 - An equation where the highest power is odd must have at least one real root.
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